Image Analysis Final Project

Breaking is Bad: Implementation of a Basic Crack Detector

By Group 26

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Data Acquisition Details

The images were taken in Germany and during trips to Delhi and Kerala in India.

Crack 1

- **Location**: Local Grocery Store in Vasundhara Enclave, Delhi, India
- **Details**: A crack in the wall near the produce section.

Crack 2

- **Location**: Apartment Complex, Vasundhara Enclave, Delhi, India
- **Details**: Crack in the parking lot.

Crack 3

- **Location**: Student Apartment in Merketalstrasse, Weimar, Germany
- **Details**: A crack in the ceiling.

Crack 4

- **Location**: Public Bathroom in a Shopping Mall in Noida, Uttar Pradesh, India
- **Details**: A crack in the wall.

Crack 5

- **Location**: Local Bus Stand in North Paravur, Kerala, India
- **Details**: A crack in the wall near the waiting area.

Crack 6

- **Location**: Paravur Majistrate Court, North Paravur, Kerala, India
- **Details**: A crack in the exterior wall of the building.

Crack 7

- **Location**: Student Apartment in Merketalstrasse, Weimar, Germany
- **Details**: A crack in the wall near the window.

Crack 8

- **Location**: Residential Property in North Paravur, Kerala, India
- **Details**: Crack in the ceiling of the storeroom.

Crack 9

- **Location**: Local Bus Stand in North Paravur, Kerala, India
- **Details**: Another crack in the outer wall of the coffeeshop near the entrance.

Crack 10

- **Location**: Petrol Pump in Edappally, Ernakulam, Kerala, India
- **Details**: Crack in the concrete wall near one of the fuel dispensers.

Dataset Statistics

Number	Crack Name	Total Pixels	Intensity Distribution Mean	Number of Labeled Pixels
1	crack1	1555840	144.7	17350
2	crack2	1008016	163.8	4217
3	crack3	602276	189.5	3431
4	crack4	1154630	159.8	3841
5	crack5	992016	151.6	6818
6	crack6	524288	118.1	6113
7	crack7	1000000	147.9	5247
8	crack8	524288	170.1	6872
9	crack9	811801	194.0	5668
10	crack10	636219	123.9	1771

(Total Images: 10)

Reasons to use a Heuristic Classifier for Crack Detection.

1. Speed

In general, heuristic approaches are faster.

2. Simplicity

Easier to comprehend and use. Area, aspect ratio, and circularity are simple qualities to calculate.

3. Flexibility

Heuristic methods allow you to create your own rules and easily adjust to new images or settings.

4. Less Data Dependent

Unlike machine learning models, which require a large amount of data for training, heuristic classifiers can function well with less data, as is the case in this example.

5. Interpretability

Each heuristic or rule has a straightforward interpretation (for example, "a crack will typically have a high aspect ratio"), which might be helpful in understanding why a particular detection was made.

The code uses a heuristic classifier to recognize cracks in a quick, simple, and adaptable manner.

Detector Performance on the Test Set (crack2.png and crack4.png)

Metrics:

- **Intersection over Union (IoU) for crack2.png**: 0.8768
- **Intersection over Union (IoU) for crack4.png**: 0.8594

Evaluation:

1. **High IoU Scores**:
    - Both 'crack2.png' and 'crack4.png' images have an IoU score greater than 0.85.
    - This indicates that the detected region has a significant overlap with the ground truth.
    - Conclusion: The detector is effective at identifying cracks.

The Metric's Suitability for Crack Detection

1. **High Sensitivity**:
    - IoU is sensitive to true positives, false positives, and false negatives.
    - Makes it suitable for object detection tasks such as crack detection.
2. **Balance of Precision and Recall**:
    - IoU provides a good trade-off between Precision and Recall.
    - Simplifies the task of model comparison.
3. **Single Value Metric**:
    - A single value like IoU is more straightforward for comparisons between multiple models or algorithms.
4. **Imperfect Ground Truth**:
    - The ground truth may contain errors.
    - High IoU scores assume the ground truth is always accurate, which might not be the case.
5. **Small Cracks**:
    - IoU may not be the best metric for detecting very small or thin cracks.
6. **Context-Awareness**:
    - IoU does not evaluate the significance of the detected cracks in the context of the overall structure.
    - Some cracks may be more critical than others due to their location or size, but IoU treats all detections equally.

Crack Lengths (Normalised)

The crack length is normalized by dividing total_length, that is, the total length of identified cracks in the input binary image, by the total number of pixels in the image.

Crack ID	Crack Length
Crack 1	3.901819e-02
Crack 2	1.557497e-02
Crack 3	2.054995e-02
Crack 4	1.214513e-02
Crack 5	2.517770e-02
Crack 6	4.308083e-02
Crack 7	1.950001e-02
Crack 8	4.870582e-02
Crack 9	2.530981e-02
Crack 10	9.407797e-03

Assessing the Practicality of the Implemented Approach for Crack Detection: Strengths and Weaknesses

Advantages

1. **In-depth Preprocessing:** The algorithm incorporates robust preprocessing processes such as Gaussian blurring, Sobel gradient, and thresholding that aim to highlight characteristics of interest (cracks) while minimizing noise.
2. **Non-Maximum Suppression:** This method improves the edge features, making it easier to distinguish between crack and non-crack regions.
3. **Morphological Operations:** To fill minor gaps and improve crack continuity, dynamic morphological operations based on average crack area are performed.
4. **Feature Development:** The code extracts various features (area, aspect ratio, and circularity) to differentiate fractures from other types of noise, making it a potentially adaptable model.
5. **Multiple Level Thresholding:** The code provides a more subtle technique to classify areas as cracks or not by setting weighted criteria based on several features.

Limitations

1. **Required Tuning:** The model includes numerous hyperparameters (e.g., thresholds, kernel sizes) that may need to be fine-tuned depending on the type of photos and fractures.
2. **Sensitivity to Noise:** Despite the fact that the code includes thresholding and morphological processes, noisy or grainy photos may create mistakes in crack detection.
3. **Complexity:** The code is lengthy and has numerous steps, making it computationally expensive, especially for real-time applications.
4. **Strictly Limited Feature Set:** While the code does perform feature extraction, it is quite simple. Advanced machine learning models may provide higher performance.

Crack Detection Practical Implications

- **Real-World Application:** This approach could be tremendously beneficial for structural health monitoring in industries such as civil engineering, manufacturing, and aerospace with further improvement and validation.
- **Applications in Industry:** With further improvement and validation, this approach could be extremely useful for structural health monitoring in industries such as civil engineering, manufacturing, and aerospace.